

116 Advanced Statistics for Neuroscience

Exercise 06

Bootstrapping and Linear Discriminant Analysis

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Overview

In this exercise, you will use bootstrapping to obtain confidence intervals and linear discriminant analysis to increase statistical power.

You are provided with observations that are 'multivariate' in the sense that every observation has x - and y -values. (The principle would be the same if every observation had 3, 10, or 100 values.)

You have two sets, 'Samples A' and 'Samples B', with 30 observations each. Your question is whether the samples come from different distributions. In other words, can you **rule out** the null-hypothesis that they come from the same distribution?

Parametric and one-dimensional analysis

The observations are provided in file **SamplesForExer7.txt**.

Visualize your data with a **scatter plot** in the x-y-plane. Use color to distinguish observations from different sets.

Test the null-hypothesis parametrically and separately for x- and y-values. Proceed as follows:

- ▶ Compute means, variances, and standard errors for each coordinate and set:

$$\mu_{x^A} = \frac{1}{30} \sum_i x_i^A, \quad \sigma_{x^A}^2 = \frac{1}{29} \sum_i \left(x_i^A - \mu_{x^A} \right)^2, \quad SE_{x^A} = \sqrt{\frac{\sigma_{x^A}^2}{30}} \dots$$

- ▶ Test the null-hypothesis that the difference between means is zero in the x- and y-direction

$$t_x = \frac{\mu_{x^A} - \mu_{x^B}}{\sqrt{SE_{x^A}^2 + SE_{x^B}^2}}, \quad p_x = 2 \cdot [1 - CDF(t_x)]$$

Parametric and two-dimensional analysis

Combine the statistical power of both coordinates with **linear discriminant analysis**. Proceed as follows:

- ▶ Compute centroids and covariance matrices

$$\mathbf{m}_A = \frac{1}{n} \sum_{\mathbf{x} \in A} \mathbf{x}, \quad \mathbf{C}_A = \frac{1}{n} \sum_{\mathbf{x} \in A} (\mathbf{x} - \mathbf{m}_1)(\mathbf{x} - \mathbf{m}_1)^T$$

- ▶ Obtain most discriminating direction $\mathbf{w}$

$$\mathbf{y} = (\mathbf{C}_A + \mathbf{C}_B)^{-1}_W (\mathbf{m}_1 - \mathbf{m}_2), \quad \mathbf{w} = \frac{\mathbf{y}}{\|\mathbf{y}\|}$$

- ▶ Obtain means and variances of projections onto $\mathbf{w}$

$$m_1 = \mathbf{m}_1^T \mathbf{w}, \quad m_2 = \mathbf{m}_2^T \mathbf{w}, \quad \sigma_1^2 = \mathbf{w}^T \mathbf{C}_1 \mathbf{w}, \quad \sigma_2^2 = \mathbf{w}^T \mathbf{C}_2 \mathbf{w}$$

- ▶ Test null-hypothesis by computing t_{xy} and p_{xy} as above.

Bootstrapping

Randomly resample each of your sets (200 times with replacement) and compute the centroids for every bootstrapped sample.

Visualize the two sets in **another scatter plot** and compute the most discriminating direction. Add this direction as a **black line** to your scatter plot.

Project the centroids to the most discriminating line and compute all pairwise differences (all 40,000!) between projections. Flip the sign in the case that most differences are negative.

Obtain the **p-value** as the fraction of pairwise differences that are negative.

Final checklist

For full credit, please submit the following results

- ▶ Scatter plot with original samples and most discriminating line.
- ▶ Scatter plot with bootstrapped centroids and most discriminating line.
- ▶ Parametric p-values for x- and y-dimensions.
- ▶ Parametric p-value for combined dimensions.
- ▶ Non-parametric p-value for combined dimensions (from bootstrapping).
- ▶ Explanation as to which p-value you consider to be most robust and informative?